



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE B

Course Code & Title: EC8702 Adhoc and Wireless Sensor Networks

Name of the Faculty member (s): Dr.K.Ragavan

Innovative Practice Description

- **Unit / Topic:** Unit IV / Network Security attacks
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Flipped Classroom
- **Justification:**

The flipped class topic was chosen because there are several attacks in each layer of Wireless Sensor Networks. The details of various network security attacks are provided in Unit 4. Since the students are exposed to the basics of attacks, I decided to tell a flipped class on 'Network Security Attacks'. Only a few attacks at various layers were mentioned in their prescribed text book, despite the fact that there are many attacks in wireless sensor networks. Students have the opportunity to search and explore more information related to specific attacks if this flipped class is used in addition to the learning materials provided by the faculty member. As a result, this activity will cover various attacks in wireless sensor networks in two periods, which is not feasible in traditional teaching. This is why this topic was chosen for the flipped class activity.

- **Time Allotted for the Activity:** 2 periods
- **Details of the Implementation:**
 - The students were informed about the flipped class activity atleast 10 days before the implementation of the activity
 - The materials like PPT, documents, videos related to the topic will be sent through mail to all the students
 - The grouping of students will also be done well in advance and each group is allotted with attacks in any one layer (10 days before implementation)
 - As a reminder, a notification regarding the activity will be informed every two days
 - The students must prepare a 10-minute presentation including the topic learned, 3 minutes for questions and answers, and a description of each member's participation.
 - The inside-class activity will be a presentation that will last two sessions. In which each group must present their selected topic to the rest of the class, who will be listening.

- Layerwise attacks in network security will be discussed with the entire class, and handouts on the issue will be distributed among the group members.
- The major goal of the flipped class activity is for each group to study attacks in one layer, and for the other members to learn about all of the attacks in various layers through presentations.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO4	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO4	3	2	2	1	1	1	3	2	1
(1 – Low 2 – Moderate 3 – High)									

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO8	PO9	PO10
	3	2	2	1	1	1
Justification for correlation	Communication networks and Wireless networks fundamentals are applied	Various security attacks are identified and analyzed	possible solutions for various attacks to solve complex engineering problems in WSN	Group activity which requires inter-personal ethics	Needs individual and team work	effective communication is required

Innovative practice	PSO1	PSO2	PSO3
	3	2	1
Justification for correlation	possible solutions can be analyzed to solve the existing network security attacks	key distribution and management schemes can be applied in embedded systems for network security attacks	Analyze network security attacks in sensor networks for communication systems

• **Images / Screenshot of the practice:**



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- The students got experience learning a topic from outside class and giving a presentation in class.
- They found it totally useful to learn about the real-time attacks on wireless sensor networks because the topic is about various network security attacks.
- The activity was designed for groups, and it included interactions and conversations that allowed participants to learn more.

❖ *Benefit of the practice:*

- Because of the group activity, there is more interaction among the peer group members.
- The students felt a new experience and it helped to know how to make good presentation and avoid stage fear.
- The students were able to focus on the main topics due to a guideline for using the learning materials.

❖ *Challenges faced in implementation:*

- Time management – Since few group of students made lengthy presentation, it was difficult to complete in 10 minutes.
- Few students had stage fear while presentation.
- In Q& A module – few students struggled to answer the questions and ask question to the peers.

References:

- ❖ Holger Karl , Andreas willig, - Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006.
- ❖ Feng Zhao, Leonidas Guibas, —Wireless Sensor Networks: an information processing approach, Elsevier publication, 2004
- ❖ I.F. Akyildiz, W. Su, Sankarasubramaniam, E. Cayirci, —Wireless sensor networks: a survey, computer networks, Elsevier, 2002, 394 - 422.



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Innovative Practice Description

- **Unit / Topic:** Unit IV / Sensor Node Hardware
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO13
- **Activity Chosen:** Three Step Interview

- **Justification:**

The three-step interview learning strategy incorporates a blend of role-playing and discussion to facilitate student learning. Collaborative learning encourages the development of skills including active listening, note-taking, and knowledge sharing. The strategy comprises of Q&A sessions and interviews, enhancing student interest and engagement, and in turn making learning more fun and enjoyable in the classroom.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

For the three-step interview, students are grouped into small groups of three. Each member in the group assumes the role of interviewer, interviewee and reporter/note-taker; and each student should get an opportunity to play each role. To help explain the process, each student will be named as A, B and C.

The interviewing process is conducted in three steps:

Steps	Interviewer	Interviewee	Reporter
Step 1	Student A	Student B	Student C
Step 2	Student C	Student A	Student B
Step 3	Student B	Student C	Student A

After the three-step interview process is completed, the students are given extra time to share and consolidate the information.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	2	1	2	2	2	1	1	1	3	3	1

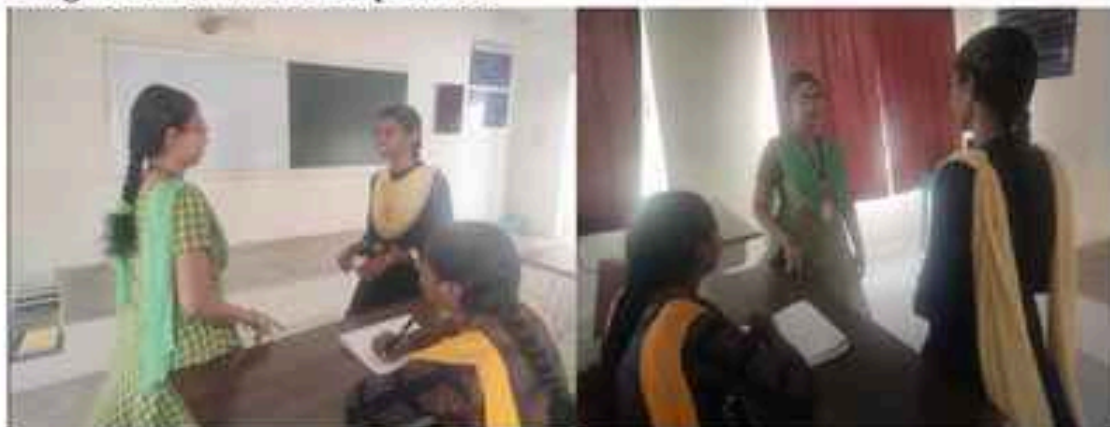
(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12
	2	1	2	2	2	1	1	1
Justification for correlation	Communication networks and Wireless networks fundamentals are applied	Various simulators are analyzed	Berkeley mote sensor hardware component are analyzed	Selection of appropriate hardware and software tools are required to design a sensor node	Node level software's are used for sensor node hardware	Needs individual and team work	effective communication is required	Sensor nodes are used in day-to-day life and it is used to learn new technology

Innovative practice	PSO1	PSO2	PSO3
	3	3	1
Justification for correlation	possible node level simulators can be analyzed to solve the existing sensor network programming challenges	Berkeley mote sensor node hardware, node level softwares can be applied in embedded systems	Sensor network platforms and tools are used for communication systems

• Images / Screenshot of the practice:





❖ **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- Most of the students enjoyed the activity because this is the first time experience
- It helps them to improve the communication skills
- It helps them familiarized with the method of interviewing

❖ ***Benefit of the practice:***

The three-step interview learning strategy is beneficial to student learning in the following ways:

- It encourages students to think about the information they have been taught
- It expands their ability to question, think and generate answers
- It provides each student with the opportunity to voice their opinions, and promotes equal participation
- It promotes student accountability

❖ ***Challenges faced in implementation:***

- Time management – Since few group of students made lengthy interview, it was difficult to complete within the time.
- few students struggled to answer the questions due to lack of preparation.

References:

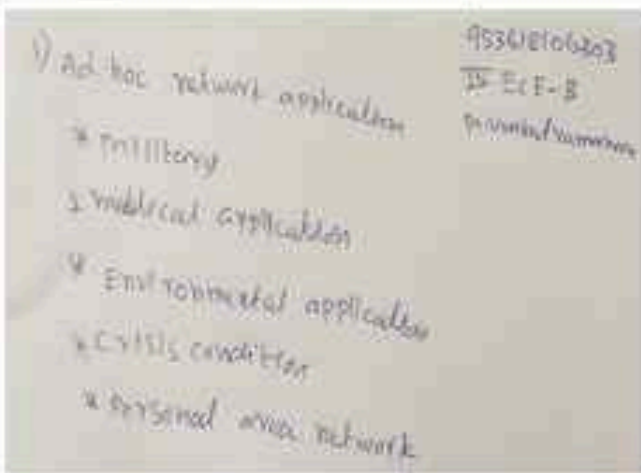
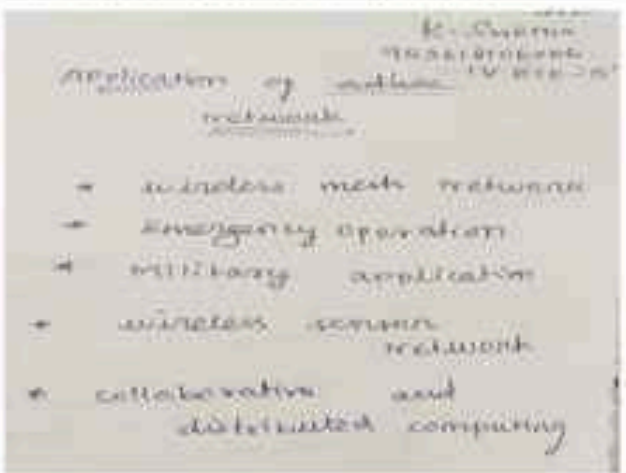
- ❖ Holger Karl , Andreas willig, - Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006.
- ❖ Feng Zhao, Leonidas Guibas, —Wireless Sensor Networks: an information processing approach, Elsevier publication, 2004
- ❖ I.F. Akyildiz, W. Su, Sankarasubramaniam, E. Cayirci, —Wireless sensor networks: a survey, computer networks, Elsevier, 2002, 394 - 422.
- ❖ Ruiz-Sandoval, Manuel; Nagayama, Tomonori; Spencer, B. F. (2006). Sensor Development Using Berkeley Mote Platform. Journal of Earthquake Engineering, 10(2), 289-309.

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INNOVATIVE TEACHING METHOD

Degree, Semester& Branch: VII Semester B.E. ECE B

Course Code & Title: EC8702 Adhoc and Wireless Sensor Networks

Name of the Faculty member: Mr.K.Ragavan

Sl.No.	Topic(s)	Activity*	Reference
UNIT I - AD HOC NETWORKS – INTRODUCTION AND ROUTING PROTOCOLS			
1.	Elements of Adhoc Wireless Networks	Class Poll	C. Siva Ram Murthy and B. S. Manoj, - Ad Hoc Wireless Networks Architectures and Protocols, Prentice Hall, PTR, 2004.
			
2.	Applications of Adhoc Networks	Exit Slip	C. Siva Ram Murthy and B. S. Manoj, - Ad Hoc Wireless Networks Architectures and Protocols, Prentice Hall, PTR, 2004.
			

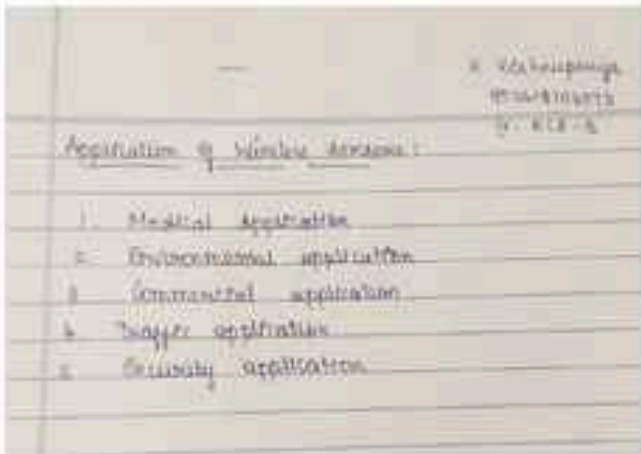
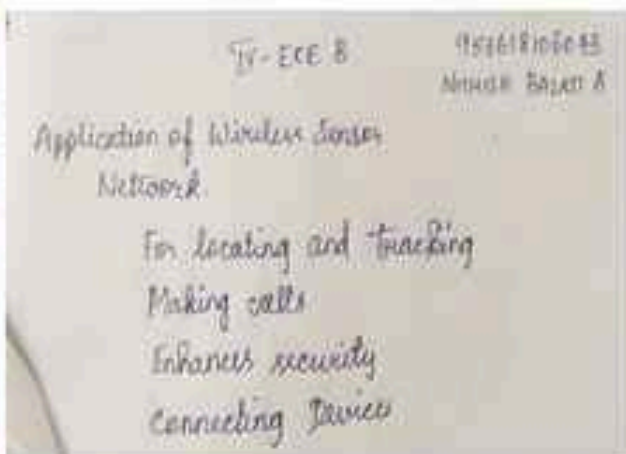
3.	Classification of Routing Protocols	Zero Minute Speech	C. Siva Ram Murthy and B. S. Manoj, - Ad Hoc Wireless Networks Architectures and Protocols, Prentice Hall, PTR, 2004,
			

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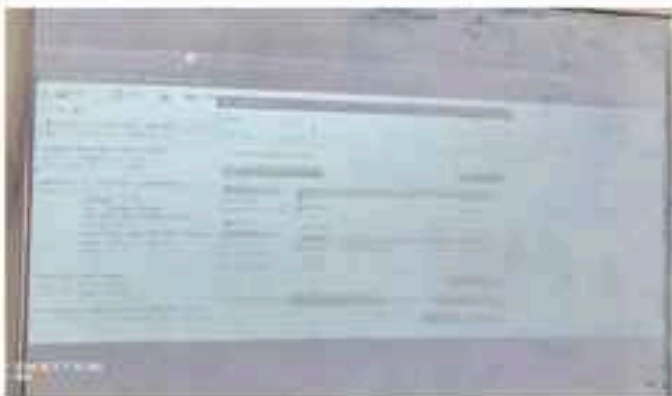
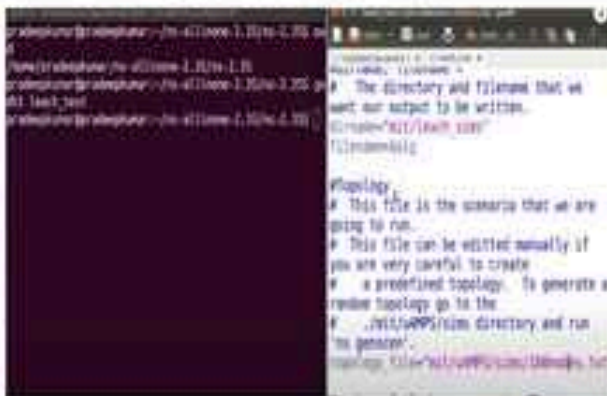
Sl.No.	Topic(s)	Activity*	Reference
UNIT II - SENSOR NETWORKS – INTRODUCTION & ARCHITECTURES			
1.	Wireless Sensor Networks - Applications	One Minute Paper	Holger Karl , Andreas willig, Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
 			
2.	Single Node Architecture- Hardware Components (Controller Part)	Hardware Demo	Holger Karl , Andreas willig, Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
 			

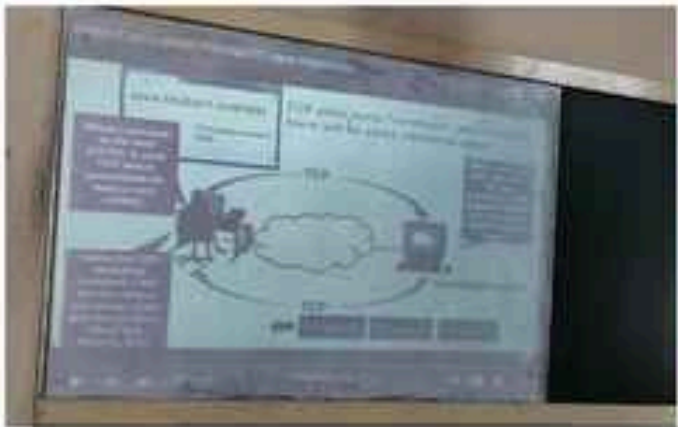
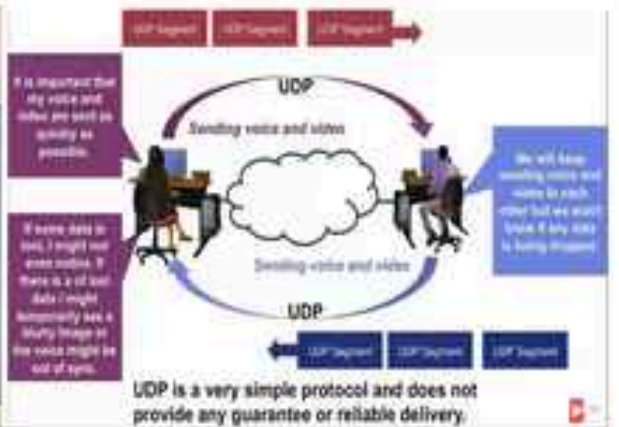
3.	Wireless Networks – applications architecture	Sensor challenges, and	Quiz Dominos	Holger Karl , Andreas willig, Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
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Sl.No.	Topic(s)	Activity*	Reference
UNIT III - WSN NETWORKING CONCEPTS AND PROTOCOLS			
1.	MAC protocols for Wireless Sensor Networks	Quescussion	Holger Karl , Andreas willig, Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
 			
2.	Schedule based Protocols-LEACH	Simulation	Holger Karl , Andreas willig, Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
 			

3.	Challenges and Issues in Transport layer Protocol	Animated Video	Holger Karl , Andreas willig, Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
			

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Sl.No.	Topic(s)	Activity*	Reference
UNIT IV - SENSOR NETWORK SECURITY			
1	Layer wise attack in Wireless Sensor Networks	Mind Map	Holger Karl , Andreas willig, Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
 			
2	Block hole attack, Flooding attack	Role Play	Holger Karl , Andreas willig, —Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
 			

3	Key Distribution and Management	Think-Pair-Share	Holger Karl , Andreas willig, —Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
			

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Sl.No.	Topic(s)	Activity*	Reference
UNIT V - SENSOR NETWORK PLATFORMS AND TOOLS			
1.	Berkeley Motes	Library	Ruiz-Sandoval, Manuel; Nagayama, Tomonori; Spencer, B. F. (2006). Sensor Development Using Berkeley Mote Platform. Journal of Earthquake Engineering, 10(2), 289–309.
			
2.	Node Level Software Platform	Pro-Con-Grid	Feng Zhao, Leonidas Guibas, —Wireless Sensor Networks: an information processing approach, Elsevier publication, 2004.
			

3.	Node Level Simulator- NS2	Simulation	Feng Zhao, Leonidas Guibas, —Wireless Sensor Networks: an information processing approach, Elsevier publication, 2004.
